LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Different errors in INS., initial leveling error, initial tilt error, accelerometer error, integrator error, initial azimuth misalignment, azimuth gyro drift, and leveling gyro drift.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Different errors in INS., initial leveling error, initial tilt error, accelerometer error, integrator error, initial azimuth misalignment, azimuth gyro drift, and leveling gyro drift.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of INS error diagrams and drift vectors; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an Inertial Navigation System?
(b) Define the term 'drift' in the context of gyroscopes.
(c) List three sources of error in an INS system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Initial leveling error definition • Initial tilt error causes • Accelerometer bias errors • Integrator error accumulation • Initial azimuth misalignment	L & Ds	PPT, IB	15 min
(b)	• Azimuth gyro drift characteristics • Leveling gyro drift mechanisms • Error correction techniques • Impact on navigation accuracy • Mitigation strategies	L & Ds	PPT, PDF	20 min
(c)	• Systematic vs random errors • Environmental error factors • Calibration procedures • Maintenance impact on errors • Operational limitations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how initial tilt error affects the calculated position.
(b) Describe the relationship between azimuth gyro drift and heading accuracy.
(c) Identify two methods to minimize integrator error in INS systems.

12. Home Assignment and Follow Up	Prepare a brief note on how Mixed inertial systems and velocity damping address the errors discussed in this lesson, linking to the next lesson on Mixed inertial systems.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Mixed inertial system, velocity damped inertial system, damping by Doppler velocity. Fix monitored inertial systems, stellar monitored inertial system.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min